Health Hazards Among E-waste Recyclers in Chattogram City, Bangladesh: A Cross- Sectional Analysis of Workplace Safety and Chronic Conditions

Maharaj Khan Muna[1], Muhammad Shahariar[2], Umme Laiba Raha[3], Dr. Alak Paul[4]

[1]Post Graduation, Department of Geography and Environmental Studies, University of Chittagong
[2]Post-graduation, Department of Geography and Environmental Studies, University of Chittagong
[3]Graduation,Department of Geography and Environmental Studies, University of Chittagong
[4]Professor, Department of Geography and Environmental Studies, University of Chittagong

Background

The rapid growth of electronic waste increases due to swift technological advancements, exposing informal recyclers in Bangladesh to hazardous substances and health risks. This study investigates safety measures, occupational health status, and issues among e-waste workers at Chittagong City Corporation (CCC) and assesses the related health problems both accidental and chronic.

Objective

- To detect association between recycling activities and different types of injuries among recyclers of e-waste.
- To assess if there any correlation between chronic diseases and heavy metal exposures extracted from e-waste.
- To observe occupational health status, healthcare practices and awareness among e-waste recyclers in Chattogram City.

Methods

Study Design: Cross Sectional (Data collected from January to March 2024 Chittagong City Corporation)

Sample Size: 84 workers

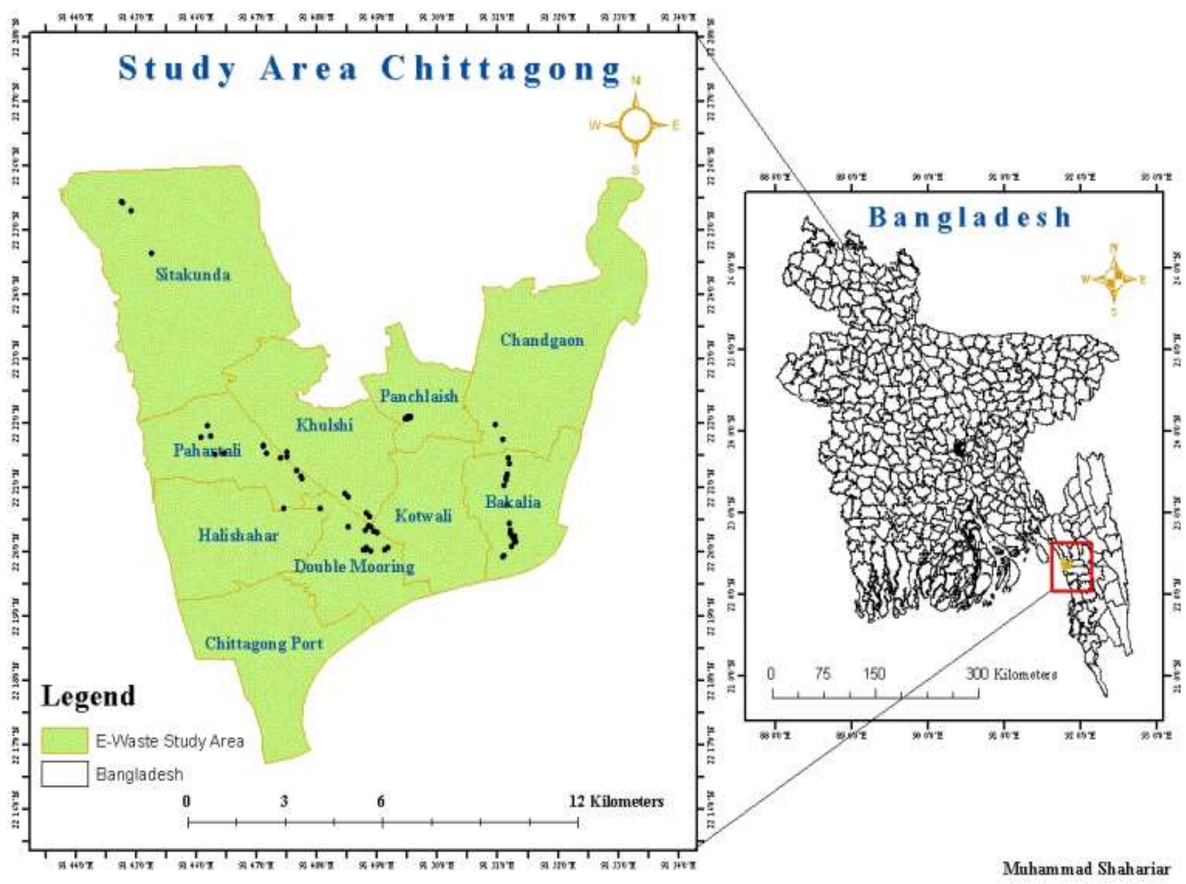
Sampling technique: Purpusive (E-waste recyclers only)

Data were collected by face to face interview

Questionnaire survey by Kobo-toolbox

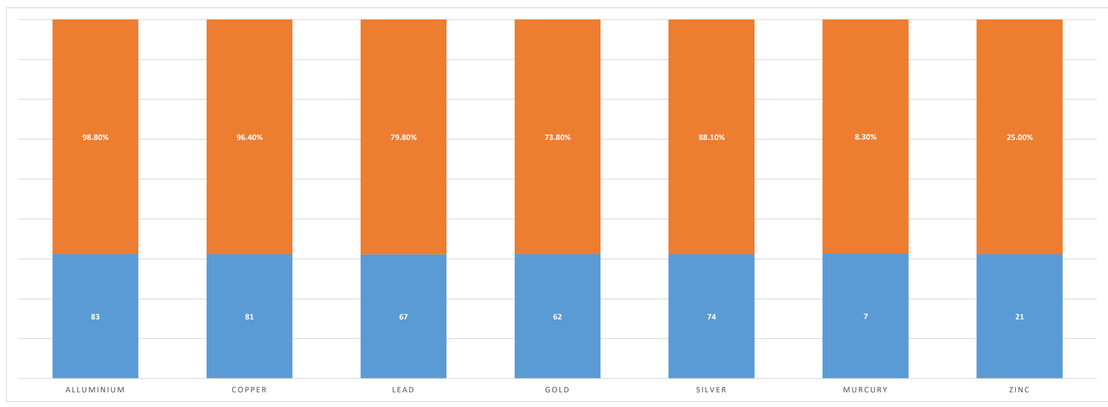
Data Analysis: By using SPSS (Statistical Package for the Social Sciences)

Fig 1: Study area map(Chattogram City)



Results

- The study found that 44% of participants were over 35 years old, all male.
- E-waste handling operations included collection (95.2%), burning (66.7%), and dismantling (72.6%), mainly generated from households (98.8%).



- Exposure to heavy metals such as aluminum (98.8%), copper (96.4%), and lead (79.8%) was common.

Injury types		Working categories								
		Collecting			Burning			Dismantling		
		Do	Don't	p-value	Do	Don't	p-value	Do	Don't	p-value
Cuts and burns	Yes	69	4	1.00	55	18	0.00	59	14	0.00
	No	11	0		1	10		2	9	
Itching and skin problems	Yes	61	4	0.57	52	13	0.00	56	9	0.00
	No	19	0		4	15		5	14	
Shortness of breath	Yes	31	2	0.64	25	8	0.23	26	7	0.33
	No	49	2		31	20		35	16	
Red and itchy eye	Yes	32	2	1.00	31	3	0.00	32	2	0.00
	No	48	2		25	25		29	21	
Back pain and neck pain	Yes	57	3	1.00	45	15	0.02	47	13	0.10
	No	23	1		11	13		14	10	
Hearing loss	Yes	2	4	1.00	2	0	0.55	2	0	1.00
	No	78	4		54	28		59	23	

Table 1: Cross-tabulation of involvement with work in e-waste management and injury

- Health issues contained cuts, burns (86.9%), skin problems (77.4%), poor ventilation, and bad smells, most workers lacked access to healthcare and safety precautions including PPE.
- A significant correlation is present among cuts and burns, skin problems, and red, itchy eyes (p = 0.00) among workers engaged in burning and dismantling.
- No association was found between heavy metal exposure and chronic illnesses such as diabetes, hypertension, depression, anxiety, and anemia.

Discussion

- Employees participating in burning and dismantling have asignificant correlation with cuts and burns, itching and skin problems, and red and itchy eyes but a small percentage of workers experience hearing issues and odor problems as they have become habituated with these problems, they said to us during the interview. According to Decharat & Kiddee, (2020) there is e-waste induces skin diseases and hearing problem is induced by e-waste and 64% had hearing and/or vision problems.
- This research showed that there is no discernible link between heavy metal exposure and the respondents' chronic illnesses (diabetes, hypertension, heart problems, depression, anxiety, and anemia). In the contrast a study report shows that 83% of those exposed to toxic substances become sick and live with long-term illness (Hossain et al., 2010).

Conclusion

It is possible to improve worker health, increase environmental sustainability in Bangladesh, and provide guidelines for future safety and health measures by encouraging proper recycling, addressing shortcomings in e-waste management, and increasing awareness.

References

- Decharat, S., & Kiddee, P. (2020). Health Problems Among Workers Who Recycle Electronic Waste in Southern Thailand. *Osong Public Health and Research Perspectives*, *11*(1), 34–43. https://doi.org/10.24171/j.phrp.2020.11.1.06
- Forti, V., Balde, C. P., Kuehr, R., & Bel, G. (2020). *The Global E-waste Monitor 2020: Quantities, flows and the circular economy potential*. United Nations University/United Nations Institute for Training and Research, International Telecommunication Union, and International Solid Waste Association. https://collections.unu.edu/view/UNU:7737
- Hasibul. (2022). *Prospect of E-waste in Bangladesh: A review*. https://doi.org/10.13140/RG.2.2.15582.51522